

Null-Model Choice for Rule-Defined Pattern Counts in Clustered Longitudinal Sequences:

A Case Study of Jewish Halachic Menstrual Anticipation Rules

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The dataset analysed in this study is the publicly archived Marquette University menstrual cycle dataset (Fehring, 2012), collected in the randomised trial of Fehring et al. (2013). The author declares no conflicts of interest. Analysis code, the filtered dataset and cached Monte Carlo replicates are available at <https://github.com/dvirross/VessetVectorTest>.

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Abstract

Counts of deterministically defined patterns in clustered longitudinal sequences can depend strongly on the exchangeability null used to define “chance”: the same excess can be generated by heterogeneity between subjects or by ordering within subjects, and the counts are typically small, integer-valued and dependent by construction. We illustrate the problem with menstrual-cycle patterns defined by Jewish religious law (*Halacha*), whose *vesset* rules require a woman to anticipate menstruation from exact arithmetic events in her history of inter-menstrual intervals — three equal consecutive intervals, a constant first or second difference, a repeated weekly anchor. We analyse 1,554 cycles from 118 women under three exchangeability nulls for the same five-count vector: a global permutation null, a multinomial i.i.d. null, and a within-woman permutation null that conditions on each woman’s multiset of cycle lengths and randomises only their order. All are evaluated with two-sided Monte Carlo marginal tests and a joint quadratic-form (Mahalanobis) permutation statistic whose covariance is estimated from the null replicates. Under the global nulls the count vector is displaced far into the tail ($D_M = 5.08$, $p < .001$; Haflaga $z = 4.50$). Under the within-woman null it is not ($D_M = 2.93$, $p = .13$; smallest Holm-adjusted marginal $p = .19$), and all five observed counts lie at or below their conditional null means. Simulation shows type-I error at or below the nominal level for the marginal and Holm procedures and approximately nominal for the joint test, quantifies the power of the within-woman tests against simulated autoregressive and exact-repetition alternatives, and shows the conclusions are insensitive to unequal follow-up. The contrast is consistent with between-woman heterogeneity in regularity generating the global excess, and provides no evidence that within-woman ordering produces additional rule-defined events; it does not establish the absence of serial dependence.

Keywords: exchangeability, Halacha, menstrual cycle, permutation test, within-subject null

1. Introduction

Menstrual cycle length varies both within and between women (Bull et al., 2019; Creinin et al., 2004; Harlow et al., 2000; Münster et al., 1992). The population distribution is bell-shaped with a pronounced right tail; within-woman consistency is moderate, with intraclass correlations typically between 0.3 and 0.5 (Dasharathy et al., 2012), and the menstrual cycle is increasingly treated as a within-person process requiring within-person modelling (Schmalenberger et al., 2021). Hierarchical state-space models exploit this structure for cycle-length prediction (Bortot et al., 2010), and recent multi-cohort work documents low-order autocorrelation in cycle lengths (Ecochard et al., 2024).

In Jewish religious law (*Halacha*) this regularity is also the object of a detailed legal system. A woman is required to anticipate the onset of her next menstruation from patterns in her prior cycle history; the anticipatory rules — the *halachic vessel rules* (*vessel*: menstruation) — govern abstinence obligations within *taharat hamishpacha*, the laws of family purity. Each rule defines an exact arithmetic event in the sequence of inter-menstrual intervals. Of the 17 rule types in the halachic literature, 10 require calendar or time-of-day data that standard cycle records do not contain; 7 are computable from interval lengths alone. Despite centuries of practical application and the availability of longitudinal cycle datasets, we are not aware of any formal randomisation-based test of whether these events occur more often than chance in empirical data (no systematic literature search was performed).

From a statistical point of view the question is a case of a general problem: testing whether counts of deterministically defined patterns in clustered longitudinal sequences exceed a null expectation, when (i) subjects contribute sequences of very unequal length (here 5–45 cycles), (ii) the counts are integer-valued and small, (iii) the counts are dependent by definition (one rule’s event set is a subset of another’s), and (iv) the same excess can be generated either by heterogeneity between subjects or by ordering within subjects. The choice of null is therefore not a technicality. Randomisation tests derive their validity from the exchangeability structure they impose (Edgington and Onghena, 2007; Good, 2005); permutation inference for longitudinal and clustered data has been studied extensively (Friedrich et al., 2017; Lee and Braun, 2012), exchangeability-block permutation is standard in neuroimaging (Nichols and Holmes, 2002;

Winkler et al., 2014), and conditional permutation tests that hold part of the data structure fixed have a modern theory (Berrett et al., 2020; Hemerik and Goeman, 2018). This paper applies these established tools to the vessel counts and uses the application to make one point concrete: for the same data, a population-level exchangeability null and a within-subject conditional null return diametrically different answers, and only the latter addresses the question the rules implicitly pose.

We compare three nulls — a global permutation null H_G , a multinomial i.i.d. null H_{iid} and a within-woman permutation null H_W — using the same two-sided Monte Carlo marginal tests and the same joint quadratic-form statistic, a Mahalanobis distance from the simulated null mean with covariance estimated from the null replicates. The joint statistic is an adaptation of well-known quadratic-form permutation tests (Anderson, 2001; Pesarin and Salmaso, 2010; Székely and Rizzo, 2013); we make no claim of methodological novelty for it and instead verify its calibration by simulation. The study extends a line of quantitative halachic scholarship (Ross, 2019, 2021, 2022a,b) and is presented as a Case Study.

2. Background

2.1 The Polynomial Hierarchy of Vessel Rules

Let $L_1, \dots, L_n$ denote one woman's cycle lengths (days) and define the *haflaga* $H_k = L_k + 1$, the interval from the first day of menstruation k to the first day of menstruation $k + 1$, following halachic convention (Ross, 2021). The five rule types analysed here are:

Haflaga. $H_k = H_{k-1} = H_{k-2}$: three or more consecutive equal haflaga values. Degree-0 (constant) sequence; minimum run 3 cycles.

Dilug. $H_k - H_{k-1} = H_{k-1} - H_{k-2} = d \neq 0$: constant non-zero first difference. Degree-1 (linear); minimum run 3.

Dilug-in-Dilug. Constant non-zero second difference, $(H_k - H_{k-1}) - (H_{k-1} - H_{k-2}) = (H_{k-1} - H_{k-2}) - (H_{k-2} - H_{k-3}) = d \neq 0$. Degree-2 (quadratic); minimum run 4.

Week. $H_k = H_{k-1} \in \mathcal{W}$, where $\mathcal{W} = \{H_{\min} + 3, H_{\min} + 10, \dots\}$ are values spaced 7 apart (same weekday at each onset). Degree-0 anchored to weekly values; minimum run 2.

Week-Dilug. $H_k = H_{k-1} = 30$. A haflaga of 30 (29 days, four weeks plus one day) is the unique interval for which each successive onset falls one weekday later, producing an arithmetic progression of weekdays — hence the name (Hebrew *shavu'a be-dilug*; Ross, 2022b). Degree-0 at the canonical anchor; minimum run 2.

Haflaga, Dilug and Dilug-in-Dilug follow increasing polynomial order in cycle index; Week and Week-Dilug are specialisations of the degree-0 case anchored to calendar values. Two further computable rules (*Dilug Chozar Chalila* and *Haflaga Chozar Chalila*, repeating multi-value cycles of intervals) require windows of at least six to nine cycles; they are discussed in Section 3.1.

2.2 Dependence Structure Implied by the Definitions

Two relationships follow from the definitions and are relevant to how the counts should be tested jointly.

Haflaga-30 $\subseteq$ Week-Dilug (deterministic containment). — Week-Dilug requires two consecutive haflaga values equal to 30; a Haflaga sequence with value 30 requires three. Every Haflaga-30 sequence therefore generates a Week-Dilug event in its first two cycles. This is set-theoretic inclusion, not a stochastic association: in the data, the 3 Haflaga sequences with $H = 30$ contribute directly to the Week-Dilug count of 18.

Haflaga and Dilug-in-Dilug (algebraic proximity). — When the second difference $|d|$ of a Dilug-in-Dilug sequence is 1, its first differences change by one day per cycle and the haflaga sequence is nearly constant. In the data, 16 of 27 Dilug-in-Dilug sequences have $|d| = 1$. This makes positive co-occurrence with Haflaga plausible but does not entail it.

A third relation, mutual exclusion of Dilug ($\Delta H \neq 0$) and Week-Dilug ($\Delta H = 0$) within the same window, implies a weak negative tendency for that pair. The five counts are thus non-independent by construction, which motivates estimating their joint null covariance rather than treating them as separate tests.

2.3 Prior Work and Dataset Overlap

Statistical characterisation of cycle-length variability is extensive (Bull et al., 2019; Creinin et al., 2004; Dasharathy et al., 2012; Fehring et al., 2006; Harlow et al., 2000; Münster et al., 1992; Symul et al., 2019), and hierarchical models for sequential prediction exist (Bortot et al., 2010). Quantitative halachic analyses have modelled cycle transitions with Markov chains (Ross, 2022a), examined the reliability of self-reported cycle data in a halachic setting (Ross, 2019), and analysed abstinence rules statistically; none tested pattern frequencies against a randomisation null.

Ecochard et al. (2024) report evidence of an internal circamonthly oscillator using multi-cohort data that include a North American cohort collected by Fehring and Schneider, two of the investigators of the trial from which the dataset analysed here was drawn (Fehring, 2012; Fehring et al., 2013). Their Marquette cohort (1,649 cycles, 159 women) and the present sample (1,554 cycles, 118 women after exclusions) are probably drawn from the same data-collection infrastructure and should not be regarded as independent samples. They document second- to fourth-order autocorrelation in cycle lengths. We note the implication for H_W precisely: within-woman permutation preserves each woman’s multiset of cycle lengths and her sequence length, and *destroys* the observed ordering and therefore any serial dependence. Autocorrelation is not preserved under H_W ; rather, H_W is the reference distribution in which it is absent, so a rejection of H_W would indicate ordering structure that produces rule-defined events, and a non-rejection indicates only that these specific count statistics are not unusual relative to random reorderings of each woman’s own values.

3. Methods

3.1 Dataset

We analysed the publicly archived longitudinal dataset of Fehring (2012), collected in a randomised trial of two Internet-supported fertility-awareness-based methods of family planning (Fehring et al., 2013); we refer to the participants as natural family planning (NFP) users. The archived file contains 1,665 cycles from 159 women. Every recorded cycle length in it lies between 18 and 54 days and none is missing, so no cycle-length criterion was applied by

us; whether the archive itself was restricted to this range by the original investigators is not documented in the materials available to us. Three women whose cycle records appeared twice in the archive (16 rows in total, each with at most three distinct cycles) were removed, women with fewer than 5 recorded cycles were excluded (38 women, 87 cycles), and for one further woman whose eight cycles appeared as two near-identical copies (differing in one recorded length) a single copy was retained. The analysed sample comprises **1,554 cycles from 118 women** ($M = 13.2$ cycles per woman, $SD = 6.1$, range 5–45). Cycle lengths ranged from 18 to 54 days ($M = 29.3$, $SD = 3.9$, $Mdn = 28.5$, skewness = 1.29); 9.6% of cycles were 35 days or longer. The source trial (ClinicalTrials.gov NCT00843336) enrolled women aged 18–42 years who reported cycle lengths of 21–42 days at entry (Fehring et al., 2013); in the analysed sample age is recorded for 99 of the 118 women and ranges from 21 to 43 years ($M = 31.8$, $SD = 5.4$). Parity was not recorded. The data come from two charting cohorts ($n = 69$ and 49 women in the filtered sample); cohort membership was not used in the analysis. All lengths were incremented by 1 to the haflaga convention, giving $M = 30.3$, $SD = 3.9$.

Because the pattern detectors operate on *consecutive* cycles, it matters whether the exclusions could have created artificial adjacency across a removed cycle. They could not: the filtering removed whole women or duplicated rows only, and in the filtered file every woman’s recorded `CycleNumber` runs $1, 2, \dots, n_i$ without gaps (verified programmatically and asserted in the analysis code), so no pattern window spans an excluded cycle. The five-cycle minimum ensures that every retained woman can in principle produce each of the five pattern types, the longest of which requires four consecutive intervals. The filtering script is part of the code release and reproduces the filtered file exactly from the archived source file (Section 6).

Of the seven rule types computable from cycle-length data, the two repeating-cycle rules (*Dilug Chozer Chalila*, *Haflaga Chozer Chalila*) had no observed occurrences. The five-pattern family analysed here was therefore selected after inspection of the observed data and should be regarded as data-informed rather than pre-specified; the two zero-occurrence rules were not implemented in the analysis code and were not included in the multivariate analysis or in the Holm family. The analyses reported below are accordingly described as primary rather than confirmatory.

As a descriptive characterisation, the 20 women with at least one Haflaga event had a mean within-woman SD of cycle length of 1.54 days (between-woman SD 0.59), compared with 3.17 days (1.75) for the 35 women with no event of any type (Mann–Whitney $U = 133$, $p < .001$; Welch $t = -5.05$, $df = 45.5$; 95% CI for the difference $[-2.29, -0.98]$ days; Cohen’s $d = -1.13$). Because Haflaga is *defined* by repeated equal intervals, a lower SD among these women is expected by construction; this comparison describes the women the rule selects and is not independent validation of the rule. Within-woman SD estimates also depend on the number of cycles observed (5–45). Figure 1 shows the distribution and per-woman variability; Figure 2 summarises the data flow.

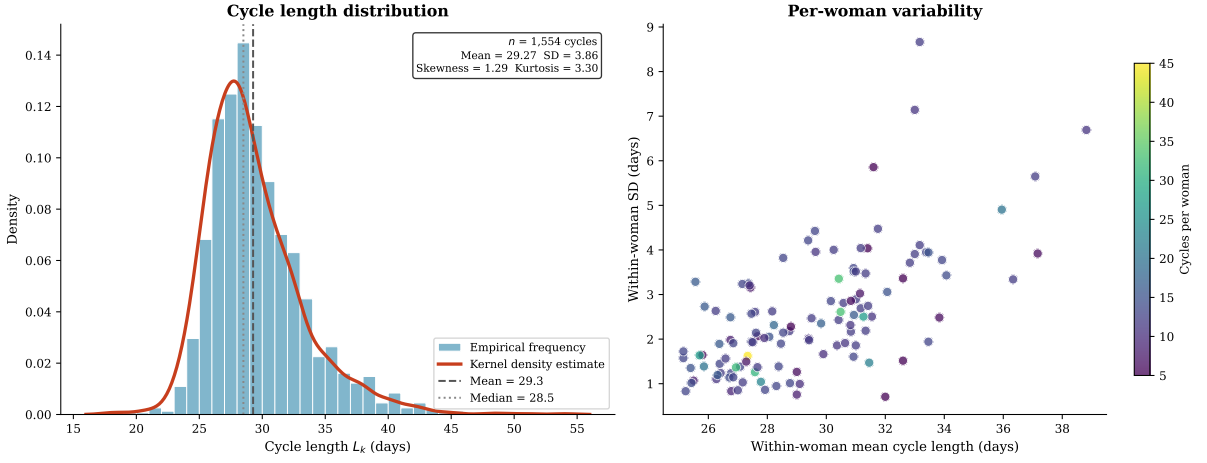


Figure 1: Cycle length distribution (filtered dataset). *Left:* histogram and kernel density estimate of L_k with mean (dashed) and median (dotted); skewness = 1.29. *Right:* per-woman mean vs. within-woman SD, shaded by cycle count.

3.2 Pattern Detection

Counting is performed within each woman’s sequence; sequences are never concatenated across women, and the woman boundaries (s_i, f_i) are enforced in every counting loop. Each function counts *distinct maximal contiguous runs* satisfying its rule: a run longer than the minimum counts once (four equal haflaga values are one Haflaga event), and a qualifying run that resumes after a discordant cycle counts again. To fix the convention: $(29, 29)$ qualifies as Week (if $29 \in \mathcal{W}$) but not as Haflaga; $(28, 28, 28)$ is a Haflaga; $(27, 29, 31, 33)$ is a Dilug ($d = 2$) with second differences $(0, 0)$ and hence not a Dilug-in-Dilug; $(27, 28, 30, 33)$ has first differences $(1, 2, 3)$ and second differences $(1, 1)$ and is a Dilug-in-Dilug.

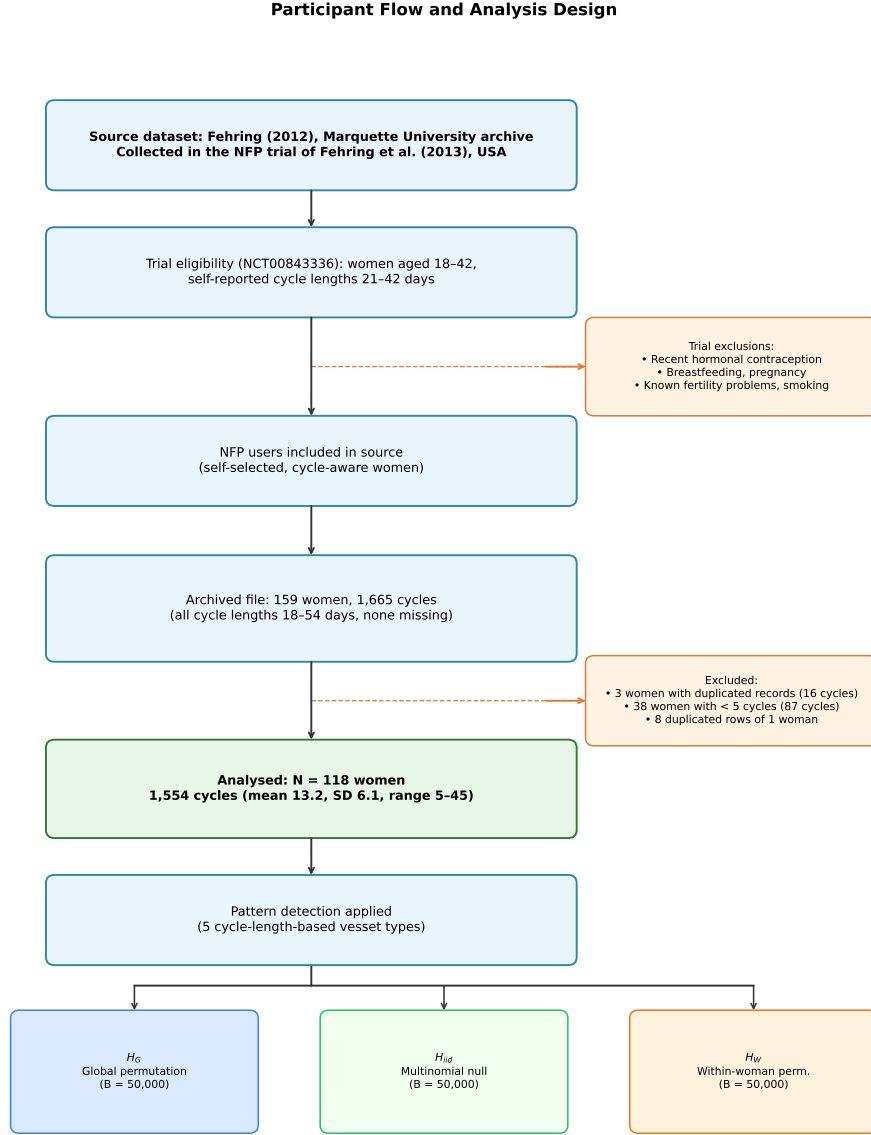


Figure 2: Participant and data flow. Source dataset (Fehring, 2012), collected in the trial of Fehring et al. (2013); archived file of 159 women and 1,665 cycles; exclusion of women with duplicated cycle records or fewer than 5 cycles; analysed sample of 118 women and 1,554 cycles; five pattern types; three null models, each with $B = 50,000$ replicates.

Two implementations were written — a reference loop version and a vectorised version used for simulation — and checked to agree on the observed data and on 9,000 random null sequences. Portions of the analysis code were generated or refactored with assistance from Claude/Claude Code (Anthropic; Claude Sonnet 4.6 and Claude Fable 5.1); all AI-assisted code was reviewed and tested by the author, and the pattern-counting code is additionally checked by this two-implementation equivalence test. The observed count vector is $\mathbf{O} = (26, 61, 32, 18, 27)$ for (Haflaga, Dilug, Week, Week-Dilug, Dilug-in-Dilug). The code, the filtered data, and all Monte Carlo replicates are available at <https://github.com/dvirross/VessetVectorTest>.

3.3 Null Models

Three nulls encode three scientific questions (Table 1).

H_G (global). All 1,554 haflaga values are exchangeable across women: the pooled values are permuted without replacement and re-assigned to women preserving each n_i . Tests whether the counts exceed what a homogeneous population would produce.

H_{iid} (multinomial). Each woman’s n_i values are drawn independently from the pooled empirical distribution $\hat{F}$ (Ross, 2022b). Stronger than H_G in that observed values, not only their ordering, are replaced.

H_W (within-woman). Conditional on each woman’s observed multiset $\{H_{i,1}, \dots, H_{i,n_i}\}$ and on n_i , all $n_i!$ orderings are equally likely. Each woman’s sequence is permuted uniformly at random without replacement (indices are permuted, so ties are handled exactly). This is a conditional Monte Carlo test (Hemerik and Goeman, 2018): it preserves each woman’s level of regularity and cycle count, destroys within-woman ordering, and tests whether the ordering generates rule-defined events beyond what her own values in random order would. It is not a general test of serial independence: an autocorrelated or trending process that does not produce additional exact events will not reject it.

A within-woman *bootstrap* (sampling each woman’s values with replacement) was considered and rejected as the conditional null: with-replacement draws are independent,

Table 1: The three null models.

	H_G (global permutation)	H_{iid} (multinomial)	H_W (within-woman permutation)
Randomised	assignment of pooled cycle values to positions	each value, drawn i.i.d. from $\hat{F}$	order of each woman's own values
Preserved	pooled multiset; each n_i	each n_i ; pooled proportions	each woman's multiset and n_i
Destroyed	woman identity; ordering	all dependence; heterogeneity	within-woman ordering only
Scientific null	counts arise from a homogeneous population	counts arise from independent draws	counts arise from random ordering of each woman's values

whereas permutation draws are negatively dependent (a value used at one position is less available at the next), so the bootstrap inflates the expected count of repetition-based patterns (Haflaga null mean 54.5 vs. 31.8; verified with $B = 50,000$). The two procedures encode different nulls — i.i.d. resampling versus exchangeability of the observed values — and are not more or less conservative versions of one another; permutation is the exact conditional test for the stated exchangeability null.

3.4 Statistical Inference

For pattern j and null m , with $B = 50,000$ replicate count vectors $\mathbf{S}^{(m,r)}$, we report the null mean $\bar{S}_j^{(m)}$ and SD $\sigma_j^{(m)}$, the standardised departure $z_j^{(m)} = (O_j - \bar{S}_j^{(m)})/\sigma_j^{(m)}$, and the two-sided Monte Carlo p -value

$$p_j^{(m)} = \min \left\{ 1, 2 \min \left(\frac{b_j^+ + 1}{B + 1}, \frac{b_j^- + 1}{B + 1} \right) \right\}, \quad (1)$$

where $b_j^+ = \#\{r : S_j^{(m,r)} \geq O_j\}$ and $b_j^- = \#\{r : S_j^{(m,r)} \leq O_j\}$, following the $(b + 1)/(B + 1)$ convention (Davison and Hinkley, 1997; Phipson and Smyth, 2010). The same convention is used everywhere (tables, figures, algorithm). Two-sided tests are used because departures in either direction are informative for these rules. Across the five marginal tests the family-wise error rate is controlled by Holm's step-down procedure, which is valid under arbitrary dependence.

The **joint test** uses the Mahalanobis distance of $\mathbf{O}$ from the simulated null mean,

$$D_M^{(m)} = \sqrt{(\mathbf{O} - \hat{\boldsymbol{\mu}}^{(m)})^\top (\hat{\boldsymbol{\Sigma}}^{(m)})^{-1} (\mathbf{O} - \hat{\boldsymbol{\mu}}^{(m)})}, \quad (2)$$

where $\hat{\mu}^{(m)}$ and $\hat{\Sigma}^{(m)} = (B-1)^{-1} \sum_r (\mathbf{S}^{(m,r)} - \bar{\mathbf{S}}^{(m)})(\mathbf{S}^{(m,r)} - \bar{\mathbf{S}}^{(m)})^\top$ are the mean and sample covariance of the null replicates (not of the observations). The p -value is the fraction of replicates whose own D_M , computed with the same $\hat{\mu}$ and $\hat{\Sigma}$, is at least the observed value, again with the $(b+1)/(B+1)$ convention. This is a single global test of the multivariate null and is inherently two-sided; it is a standard permutation quadratic form (Anderson, 2001; Pesarin and Salmaso, 2010) closely related to Hotelling’s T^2 with the parametric reference distribution replaced by a Monte Carlo one, and to energy-distance tests (Székely and Rizzo, 2013). Because the same replicates estimate $\hat{\Sigma}$ and form the reference distribution, we also report a **split-batch** version ($\hat{\mu}$, $\hat{\Sigma}$ from the first 25,000 replicates; reference distribution from the last 25,000) and verify calibration by simulation (Section 3.5). Condition numbers of $\hat{\Sigma}$ (4.4–5.4) indicate a numerically stable inversion; they say nothing about power. The joint test and the Holm family answer different questions (one global null versus family-wise error control over five component nulls) and are reported as complements.

The joint test on all five patterns is the primary multivariate analysis. An **exploratory** joint test on the three structurally linked patterns (Haflaga, Week-Dilug, Dilug-in-Dilug; Section 2.2) is reported to visualise the displacement geometry in three dimensions (Figure 8); the restriction was not pre-registered and the three-pattern result is not used for inference.

Exploratory pairwise dependence. — As a secondary consistency check against the relationships of Section 2.2, Spearman correlations between per-woman pattern *rates* (events per cycle) were computed for the ten pattern pairs, with two-sided permutation p -values and Holm correction over the ten pairs, and repeated on binary ever/never indicators. Rates reduce but do not remove the dependence of both variables on follow-up length. The methods, their rationale, the results and a follow-up-stratified robustness check are given in the Supporting Information (Section S1); they do not bear on the primary question of whether within-woman ordering produces excess events.

3.5 Simulation Validation and Sensitivity Analyses

Three questions were addressed by simulation, all using the counting code and test procedures of the main analysis (Algorithm 1).

Size. — For each null, fresh pseudo-observed datasets (10,000) were drawn from the null and analysed against an independent reference batch of 20,000 replicates. We report the rejection rate at $\alpha = .05$ of each two-sided marginal test, of the Holm family (any rejection), and of the joint test in its plug-in form (reference D_M distribution and $\hat{\Sigma}$ from the same batch, as in the main analysis) and split-batch form.

Power of the within-woman tests. — Datasets with genuine within-woman temporal structure were generated in two families and each analysed with its own 2,000 within-woman permutations (400 datasets per scenario). *AR(1)*: each woman’s sequence is a Gaussian first-order autoregressive process with her observed mean and SD and coefficient $\rho \in \{0, .25, .5, .75\}$, rounded to whole days and clipped to the observed range. *Persistence*: each cycle exactly repeats the previous haflaga with probability $q \in \{0, .1, .2, .3\}$ and is otherwise drawn from the woman’s own observed values; this is the mechanism the Haflaga rule presupposes. In both families the zero level is exchangeable within woman and serves as a further size check under heterogeneous, realistic marginals.

Sensitivity to unequal follow-up. — The H_W analysis was repeated with every woman truncated to her first 12 recorded cycles (20,000 replicates), and a leave-one-woman-out analysis recomputed the H_W joint and marginal p -values 118 times using per-woman replicate counts.

Algorithm 1 Simulation pipeline

Require: Sequences $\{(H_{i,1}, \dots, H_{i,n_i})\}_{i=1}^N$, patterns $j = 1, \dots, 5$, replicates B , seed

- 1: $\mathbf{O} \leftarrow \text{Count}(\mathbf{H})$
- 2: **for** $r = 1$ to B **do**
- 3: $\mathbf{S}^{(G,r)} \leftarrow \text{Count}(\text{Permute}(\bigcup_i \{H_{i,k}\}))$ $\triangleright$ pooled values re-assigned preserving n_i
- 4: $\mathbf{S}^{(M,r)} \leftarrow \text{Count}(\text{i.i.d. draws from } \hat{F}, n_i \text{ per woman})$
- 5: **for** $i = 1$ to N **do** $\mathbf{H}_i^{(W)} \leftarrow \text{Permute}(H_{i,1}, \dots, H_{i,n_i})$
- 6: **end for**
- 7: $\mathbf{S}^{(W,r)} \leftarrow \text{Count}(\mathbf{H}^{(W)})$
- 8: **end for**
- 9: **for** $m \in \{G, M, W\}$ **do**
- 10: $\hat{\boldsymbol{\mu}}^{(m)} \leftarrow B^{-1} \sum_r \mathbf{S}^{(m,r)}$; $\hat{\Sigma}^{(m)} \leftarrow \text{SampleCov}(\{\mathbf{S}^{(m,r)}\})$
- 11: **for** $j = 1$ to 5 **do**
- 12: $b_j^+ \leftarrow |\{r : S_j^{(m,r)} \geq O_j\}|$; $b_j^- \leftarrow |\{r : S_j^{(m,r)} \leq O_j\}|$
- 13: $p_j^{(m)} \leftarrow \min\{1, 2 \min((b_j^+ + 1)/(B + 1), (b_j^- + 1)/(B + 1))\}$ $\triangleright$ two-sided
- 14: **end for**
- 15: $D_M^{(m)} \leftarrow \sqrt{(\mathbf{O} - \hat{\boldsymbol{\mu}}^{(m)})^\top (\hat{\Sigma}^{(m)})^{-1} (\mathbf{O} - \hat{\boldsymbol{\mu}}^{(m)})}$
- 16: $D_M^{(m,r)} \leftarrow$ same with $\mathbf{S}^{(m,r)}$ in place of $\mathbf{O}$, for all r
- 17: $p_{D_M}^{(m)} \leftarrow (|\{r : D_M^{(m,r)} \geq D_M^{(m)}\}| + 1)/(B + 1)$
- 18: **end for**

4. Results

4.1 Global Nulls: Marginal Tests

Table 2 gives the marginal results under H_G and H_{iid} ; Figures 3 and 4 show the null distributions and standardised departures. Haflaga is far in the upper tail: 26 observed against a null mean of 11.31 ($z = 4.50$); only 4 of 50,000 H_G replicates reached 26 (two-sided $p = .0002$; Holm-adjusted $p = .001$). No other pattern is significant. Dilug ($z = 1.84$, $p = .085$, Holm $p = .34$) is at most hypothesis-generating; Week, Week-Dilug and Dilug-in-Dilug lie within their null ranges. The two global nulls agree closely (Figure 5; largest difference in two-sided p -values 0.11, for Week, whose null SD is larger under H_{iid}).

4.2 Within-Woman Null

The global nulls remove between-woman heterogeneity in regularity: they ask whether the counts exceed what a homogeneous population would produce, not whether each woman's *ordering* produces events beyond what her own values predict. H_W asks the latter. Table 3 and Figure 6 give the results.

Table 2: Marginal results under the global nulls ($B = 50,000$). Null mean and SD are from H_G ; z is the standardised departure under H_G ; p -values are two-sided (Eq. 1); Holm adjustment is across the five patterns within each null. Bold: Holm-adjusted $p < .05$.

Pattern	Obs.	Null mean	Null SD	z	$p(H_G)$	Holm (H_G)	$p(H_{iid})$
Haflaga	26	11.31	3.27	+4.50	< .001	.001	< .001
Dilug	61	48.92	6.57	+1.84	.085	.340	.082
Week	32	27.02	4.01	+1.24	.265	.794	.373
Week-Dilug	18	16.25	3.25	+0.54	.688	1.00	.743
Dilug-in-Dilug	27	27.17	5.00	−0.03	1.00	1.00	1.00

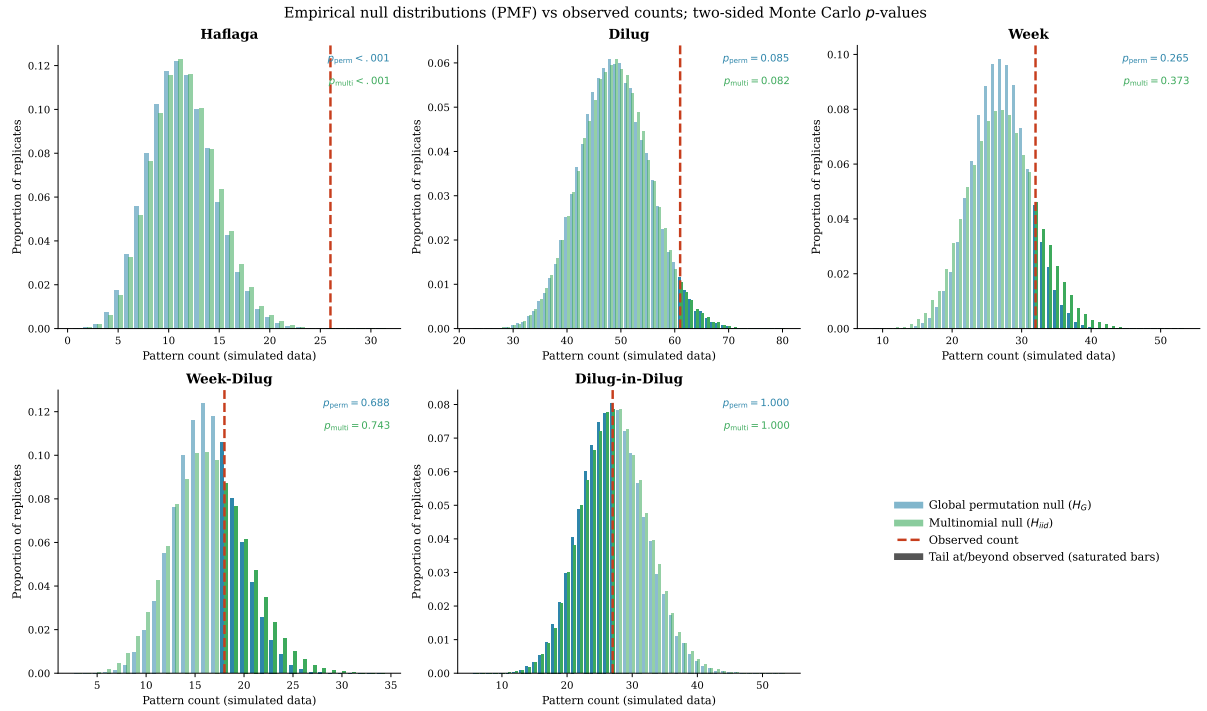


Figure 3: Empirical null probability mass functions under H_G (left bar of each pair) and H_{iid} (right bar) with the observed count (dashed line). Saturated bars mark the tail at or beyond the observed count on the observed side; the two-sided p -value is twice the smaller tail probability (Eq. 1) and is printed in each panel for both nulls.

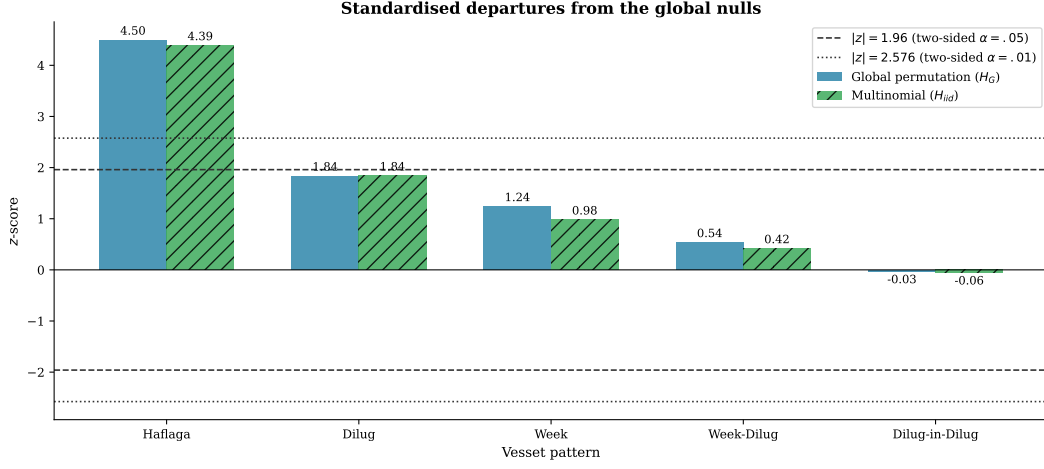


Figure 4: Standardised departures z_j under H_G (solid) and H_{iiid} (hatched). Dashed and dotted lines mark $|z| = 1.96$ and 2.576 . The two nulls give different z for the same count because each has its own mean and SD (e.g. Week: 1.24 vs. 0.98).

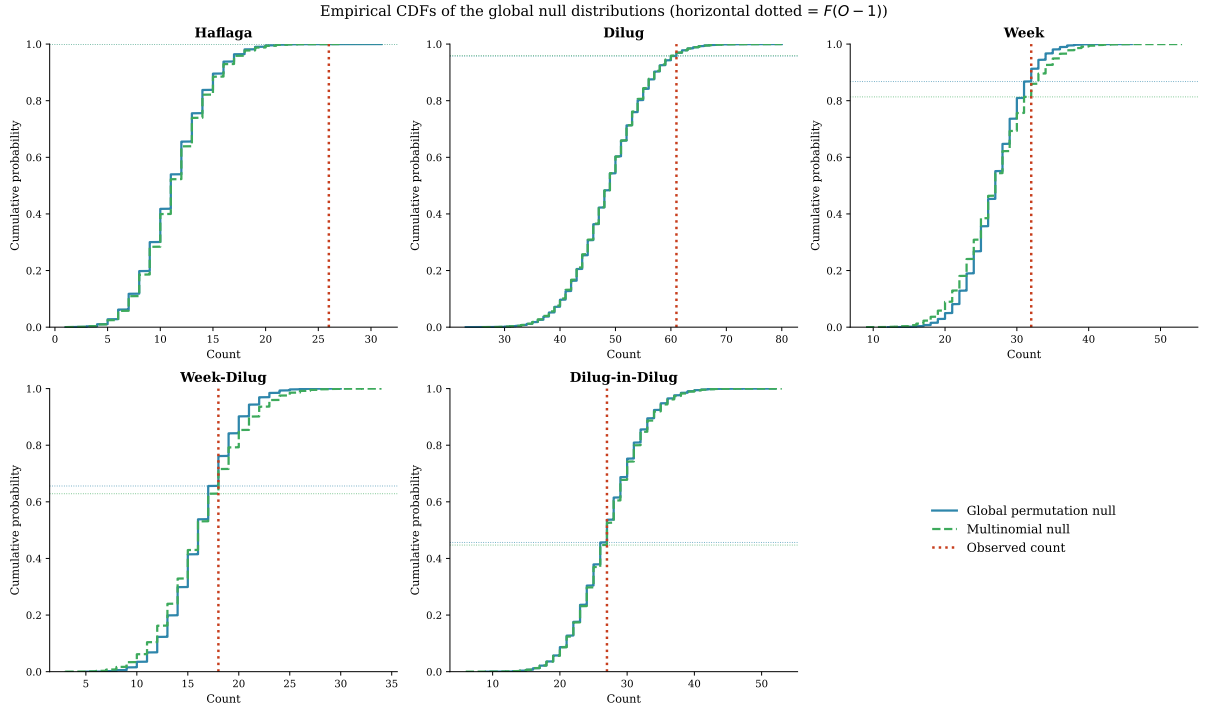


Figure 5: Empirical CDFs of the H_G (solid) and H_{iiid} (dashed) null distributions; dotted vertical line at the observed count.

Table 3: Within-woman permutation null H_W ($B = 50,000$; all 118 women; each woman’s sequence permuted without replacement). p -values are two-sided (Eq. 1); Holm adjustment across the five patterns. LOO range: range of the two-sided p -value over the 118 leave-one-woman-out analyses.

Pattern	Obs.	Null mean	Null SD	z	p	Holm	LOO range
Haflaga	26	31.84	4.68	−1.25	.254	1.00	.12–.39
Dilug	61	67.04	7.38	−0.82	.460	1.00	.34–.60
Week	32	33.79	3.88	−0.46	.749	1.00	.53–.93
Week-Dilug	18	20.91	3.22	−0.90	.460	1.00	.31–.69
Dilug-in-Dilug	27	39.12	5.84	−2.07	.038	.192	.014–.073

Joint test (five patterns): $D_M = 2.93$, $p = .126$ (6,285 of 50,000 replicates at or beyond the observed value); split-batch $D_M = 2.95$, $p = .119$; leave-one-woman-out range of p : .055–.224.

Under H_W no pattern is significant after Holm correction, and all five observed counts lie *below* their conditional null means. The most extreme component is Dilug-in-Dilug ($z = -2.07$; nominal two-sided $p = .038$, Holm $p = .19$): fewer second-order progressions were observed than random reorderings of the same values produce. The direction is noted; the mechanism is not identified by these data, and the result is exploratory. The joint test is non-significant ($D_M = 2.93$, $p = .13$) and essentially unchanged in the split-batch form ($D_M = 2.95$, $p = .12$).

The shift between nulls is large: for Haflaga the H_G null is centred at 11.3 and the H_W null at 31.8, above the observed 26. A principal contributor to this shift is that women with narrow cycle-length multisets generate many exact repetitions under *any* ordering of their values; conditioning on each woman’s multiset preserves this feature, whereas the global nulls erase it.

4.3 Joint Tests

Under H_G the five-pattern Mahalanobis distance is $D_M = 5.08$ ($p < .001$; 6 of 50,000 replicates at or beyond) and under H_{iid} it is $D_M = 4.87$ ($p < .001$; 17 of 50,000); the split-batch values are 5.07 and 4.87 with the same p -values. Under H_W , $D_M = 2.93$, $p = .126$ (Table 3; Figure 7). The displacement under the global nulls is driven by Haflaga, which has the smallest null SD (3.27) and the largest standardised departure and therefore receives the largest weight in the quadratic form; the null covariance matrices are well conditioned (condition numbers 4.6, 4.4 and 5.4 for H_G , H_{iid} , H_W) with all pairwise null correlations below 0.17 in absolute value.

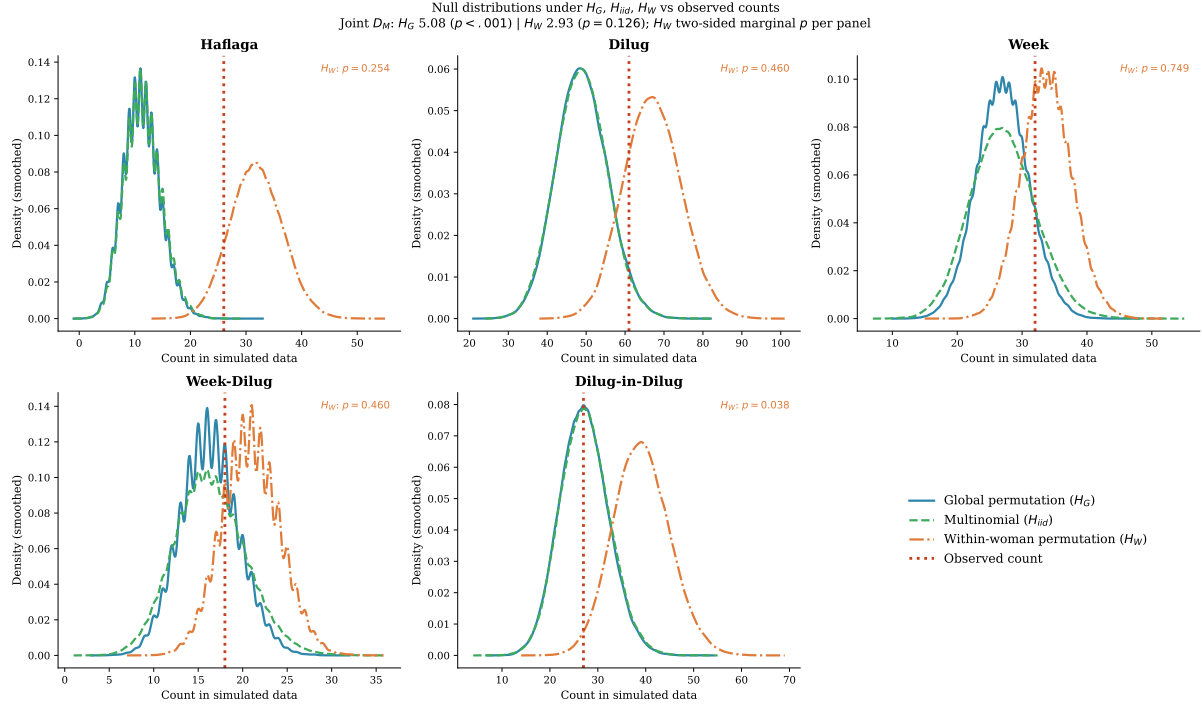


Figure 6: Smoothed null densities (for visual comparison only; inference uses the empirical PMFs) under H_G (solid), H_{iid} (dashed) and H_W (dash-dotted) for each pattern, with the observed count (dotted vertical line) and the two-sided H_W p -value in each panel.

The exploratory three-pattern test (Haflaga, Week-Dilug, Dilug-in-Dilug) gives $D_M = 4.50$ ($p < .001$), 4.39 ($p < .001$) and 2.68 ($p = .067$) under H_G , H_{iid} and H_W respectively, consistent with the five-pattern analysis (Figure 8).

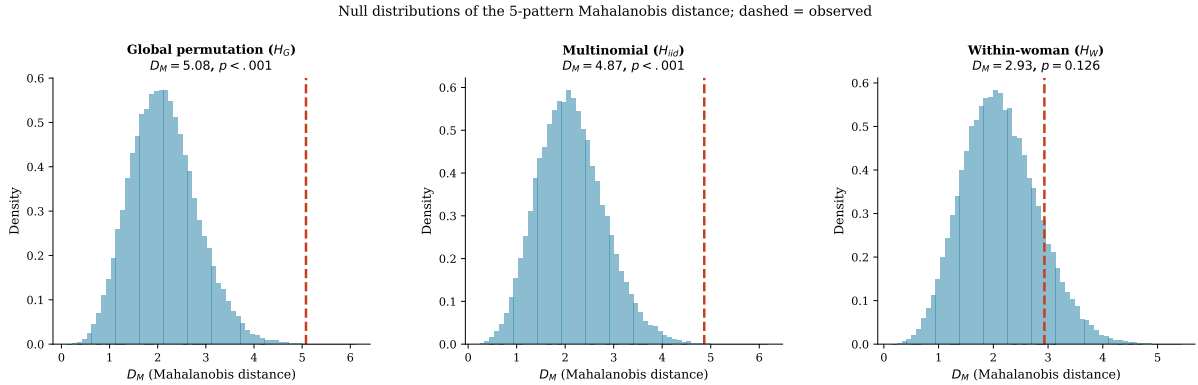


Figure 7: Null distributions of the five-pattern Mahalanobis distance D_M (Eq. 2) under H_G (left), H_{iid} (centre) and H_W (right); dashed line at the observed value.

4.4 Pairwise Dependence (Exploratory; Supporting Information)

The exploratory between-woman pairwise analysis (Supporting Information, Section S1, Table S1, Figure S1) is consistent with the relationships of Section 2.2: of the ten pairs, only

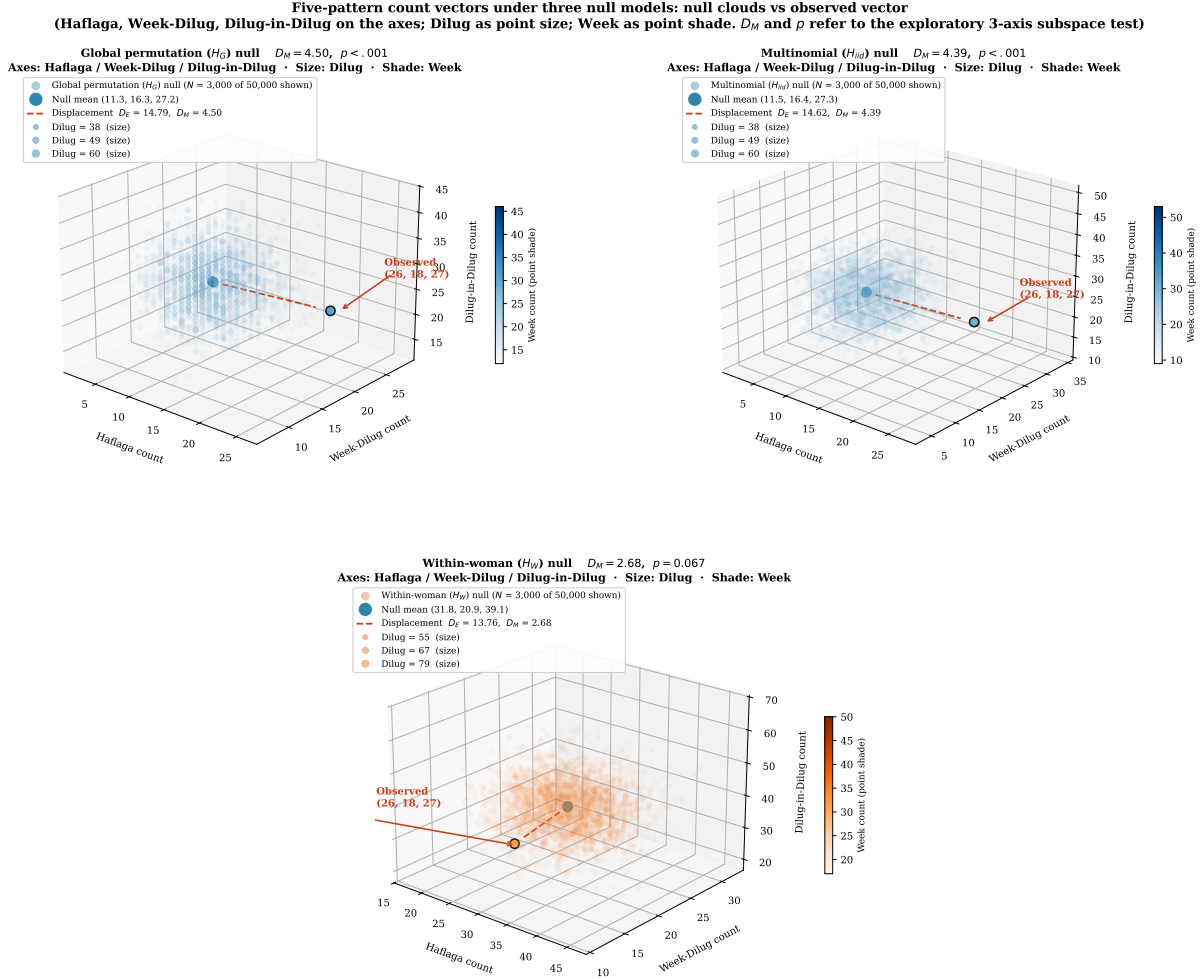


Figure 8: Five-pattern count vectors under the three null models: 3,000 of the 50,000 null replicates (points) and the observed vector under H_G (top left), H_{iid} (top right) and H_W (bottom). All five counts are shown for every replicate: Haflaga, Week-Dilug and Dilug-in-Dilug on the axes, Dilug as marker size (size legend in each panel) and Week as marker shade (colour bar). The filled circle is the null mean and the dashed segment the displacement to the observed vector (26, 18, 27) on the three axes; the D_E and D_M values in the legends and titles are those of the exploratory three-axis subspace test (Section 4.3). Primary inference uses the five-pattern D_M (Table 3).

Haflaga $\times$ Dilug-in-Dilug survives Holm correction ($r_s = 0.27$, permutation $p = .004$, Holm $p = .038$), Haflaga $\times$ Week-Dilug is positive as containment requires but does not survive ($r_s = 0.23$, Holm $p = .10$), and the remaining eight pairs are null. The binary ever/never version and a permutation stratified by follow-up length give the same ordering and essentially the same p -values. These associations are between women, not within sequences; they say nothing about temporal ordering and are compatible with, not evidence for, the proximity argument.

4.5 Simulation Validation and Sensitivity

Size. — Table 4 reports empirical rejection rates under each null for 10,000 fresh pseudo-observed datasets (Monte Carlo standard error about 0.002 at $\alpha = .05$). Type-I error is controlled at or below the nominal level by the marginal and Holm procedures and is approximately nominal for the joint test: the two-sided marginal tests reject between 2.9% and 4.8% of the time (conservative because the counts are discrete), the Holm family between 3.2% and 3.7%, and the joint test between 4.8% and 5.3% at $\alpha = .05$ and between 0.7% and 1.1% at $\alpha = .01$. The plug-in and split-batch versions of the joint test are indistinguishable, so re-using the replicates for $\hat{\Sigma}$ has no detectable effect at $B \geq 20,000$.

Table 4: Empirical size at nominal $\alpha = .05$ (10,000 pseudo-observed datasets per null, each tested against an independent batch of 20,000 replicates). Joint test shown in plug-in form ($\hat{\Sigma}$ and reference distribution from the same batch) and split-batch form; values at $\alpha = .01$ in parentheses.

Null	Marginal two-sided tests					Holm family	Joint D_M	
	Haf.	Dilug	Week	W-D	DiD		plug-in	split-batch
H_G	.029	.047	.034	.031	.046	.033	.048 (.011)	.052 (.010)
H_{iid}	.048	.038	.040	.038	.046	.032	.049 (.007)	.047 (.007)
H_W	.043	.041	.043	.042	.040	.037	.053 (.009)	.053 (.010)

Power of the within-woman tests. — Table 5 and Figure 9 give rejection rates of the H_W tests for datasets with genuine within-woman ordering (400 datasets per scenario; Monte Carlo $SE \leq 0.025$). At $\rho = 0$ and $q = 0$, which are exchangeable within woman, all tests reject at nominal rates under these heterogeneous marginals (joint 6.2%, SE 1.2%). Autoregressive dependence with $\rho = 0.25$ — which raises the expected Haflaga count by about five events over its conditional null mean — is detected by the joint test in 51% of datasets and by the Holm

family in 26%; at $\rho = 0.5$ the joint test rejects in 99.5% and at $\rho = 0.75$ always. Exact-repetition persistence is detected far more readily: when one cycle in ten copies its predecessor ($q = 0.1$) the joint test rejects in 98% of datasets and the Haflaga marginal test in 96%; at $q \geq 0.2$ detection is certain. The Dilug-in-Dilug marginal has no power against persistence (repetition suppresses second-order progressions), and Week-Dilug has the least power against autoregression, as expected from their definitions.

Table 5: Power of the H_W tests at $\alpha = .05$ against within-woman temporal-dependence alternatives (400 datasets per scenario, each analysed with its own 2,000 within-woman permutations). $\Delta\text{Haf.}$: mean observed Haflaga count minus its mean conditional null expectation. The $\rho = 0$ and $q = 0$ rows are exchangeable within woman and serve as size checks.

Alternative	Level	$\Delta\text{Haf.}$	Haf.	Dilug	Week	W-D	DiD	Holm	Joint
AR(1) ρ	0	-0.1	.04	.05	.04	.04	.04	.03	.06
	0.25	+5.1	.18	.24	.10	.06	.18	.26	.51
	0.50	+13.8	.76	.74	.35	.31	.44	.89	.995
	0.75	+33.3	1.00	.97	.78	.66	.73	1.00	1.00
Persistence q	0	+0.2	.04	.04	.04	.04	.06	.05	.06
	0.1	+24.1	.96	.19	.61	.41	.04	.94	.98
	0.2	+52.7	1.00	.60	.96	.86	.04	1.00	1.00
	0.3	+80.4	1.00	.93	.99	.96	.16	1.00	1.00

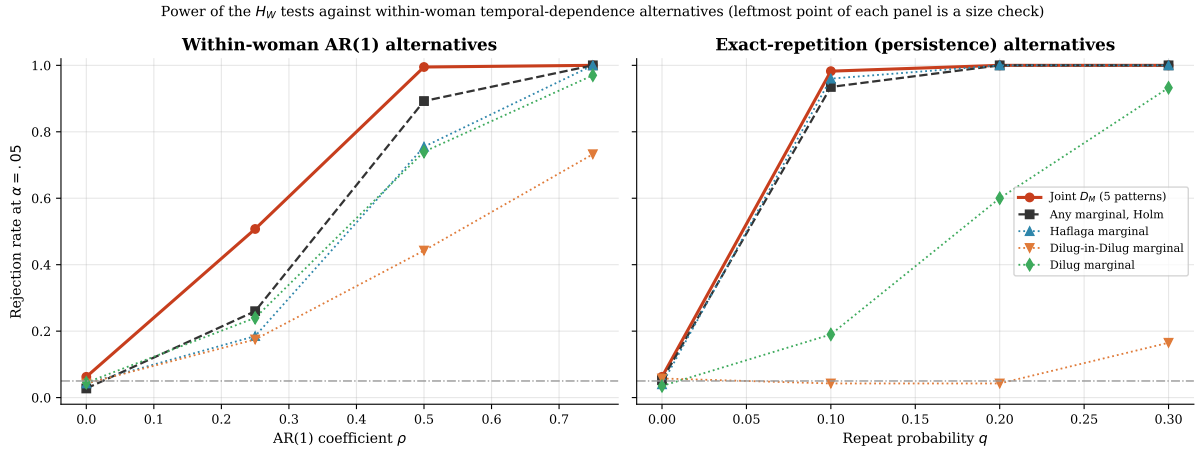


Figure 9: Rejection rates of the H_W tests at $\alpha = .05$ against within-woman AR(1) alternatives (left) and exact-repetition alternatives (right); the leftmost point of each panel is a size check. Solid: joint five-pattern D_M ; dashed: any Holm-corrected marginal; dotted: selected marginal tests. Dash-dotted horizontal line at 0.05.

These results calibrate the main finding for the two mechanisms simulated. Under the AR(1) alternative at $\rho = 0.25$, which added about five Haflaga events on average, the joint test rejected in 51% of datasets; at $\rho = 0.5$ (about 14 additional events) it rejected in 99.5%. Under

the persistence alternative, one exact repetition in ten cycles was detected in 98% of datasets. The observed Haflaga count is six *below* its conditional null mean. These figures do not transfer to arbitrary forms of temporal dependence: autoregression weaker than $\rho \approx 0.25$, or dependence that does not manifest as whole-day autoregression or exact repetition, may go undetected by these statistics.

Sensitivity to unequal follow-up. — Eleven women with 20 or more cycles contribute 20% of cycles and 35% of Haflaga events (21%, 28%, 11% and 30% of Dilug, Week, Week-Dilug and Dilug-in-Dilug events); the Spearman correlation between a woman’s cycle count and her Haflaga events is 0.28. When every woman is truncated to her first 12 cycles (1,289 cycles), the observed counts are (19, 51, 21, 16, 21), all z remain negative ($-1.18, -0.32, -0.89, -0.68, -1.78$), no marginal test approaches significance (smallest two-sided $p = .080$, for Dilug-in-Dilug), and the joint test gives $D_M = 2.49, p = .29$. Leaving out any single woman moves the joint H_W p -value within .055–.224 and never below .05 (Table 3); the nominal Dilug-in-Dilug marginal stays below .05 in 114 of 118 leave-one-out analyses, so it is not driven by one woman, but it remains non-significant after Holm correction throughout.

5. Discussion

5.1 What the Contrast Between Nulls Shows

The same five counts are extreme under population-level exchangeability ($D_M = 5.08, p < .001$) and unremarkable under within-woman exchangeability ($D_M = 2.93, p = .13$). The global result is real and is not an artefact of the multinomial versus permutation choice (the two agree). It says that vessel events are concentrated in a way a homogeneous population would not produce. The within-woman result says that, conditional on each woman’s own cycle-length values, the order in which they occurred produced no detectable excess of rule-defined events — for these exact statistics, against random reorderings, at the sensitivity documented in Section 4.5. The simplest account consistent with both is between-woman heterogeneity in regularity: women with narrow cycle-length distributions generate many exact repetitions under any ordering, and the global nulls, which pool across women, treat this as excess. This account is supported by the data; it is not uniquely identified by them, and other mechanisms that leave

the multiset-conditional distribution of these counts unchanged cannot be excluded.

Two things the H_W result does not show should be stated. It does not show that cycle lengths are serially independent: the autocorrelation documented by [Ecochard et al. \(2024\)](#) in an overlapping cohort is exactly the kind of structure that H_W destroys, and its non-detection here means only that it does not translate into extra exact vessel events in this sample. And it does not show that the rules are without anticipatory value: whether a woman with three equal consecutive intervals is more likely to repeat the interval is a forecasting question, not addressed here.

5.2 Primary, Exploratory and Robustness Analyses

Primary: the five-pattern joint tests under H_G and H_W , and the Holm-corrected marginal tests under each. *Exploratory:* the Dilug marginal ($p = .085$), the Dilug-in-Dilug deficit under H_W (Holm $p = .19$), the pairwise associations (Supporting Information), and the three-pattern subspace. *Robustness:* the multinomial null, the split-batch joint test, truncation, and leave-one-woman-out. No analysis was pre-registered; the five-pattern family was fixed after two rules were found to have zero occurrences (Section 3.1).

5.3 Pattern-Specific Findings

Haflaga is the only pattern that departs from a global null after multiplicity control and it does so strongly (4 of 50,000 replicates reached the observed count). Under H_W its count is below the conditional mean. The 20 women with Haflaga events have markedly lower within-woman SD than women with no events, as the definition of the rule implies; this characterises whom the rule selects rather than validating it. Dilug's global $z = 1.84$ does not survive correction and should be treated as hypothesis-generating; gradual trends (age, weight, hormonal change) would produce arithmetic progressions, but no such mechanism is tested here. The Week pattern in this sample is almost entirely the 28-day cycle (haflaga 29) repeated twice, because longer weekly anchors (35, 42 days) are rare (9.6% of cycles are ≥ 35 days). Week-Dilug events include the 3 Haflaga-30 sequences by containment.

5.4 The Joint Statistic

The quadratic form standardises each component by its null variance and accounts for the null covariance implied by the definitions; Haflaga dominates under the global nulls because its null SD is smallest. Hotelling’s T^2 shares the form but assumes multivariate normality of the count vector, which fails for counts as small as 18; the Monte Carlo reference distribution avoids this. A max- T statistic would be less sensitive to simultaneous displacement of several components; in these data it leads to the same conclusions under all nulls. Using the same replicates to estimate $\hat{\Sigma}$ and to form the reference distribution introduces a finite- B dependence; the split-batch results and the size simulation show it is immaterial at $B = 50,000$. We regard the statistic as a routine adaptation of permutation MANOVA-type tests ([Anderson, 2001](#)) to rule-defined counts and claim nothing further for it.

5.5 Implications

All implications are hypotheses for future study; none is tested here. A woman with an established Haflaga has a within-woman SD below about 1.6 days in this dataset, which is the situation in which calendar-based fertility awareness is most reliable ([Bortot et al., 2010](#)); whether the halachic marker adds to statistical cycle-length prediction is the subject of a companion forecasting study ([Ross, in preparation](#)). For observant Jewish women the within-woman result is consistent with the traditional understanding that the rules codify observed regularity; whether the empirical findings bear on the legal standing of the rules is outside the scope of statistics. Changes in vessel status could in principle signal a change in regularity, a clinical hypothesis that would require datasets with clinical endpoints.

5.6 Transferable Lesson

The reusable content of this case is not the statistic but the null-model comparison. Whenever deterministic rules are applied to recurrent-event sequences from many subjects with unequal follow-up — exact inter-seizure intervals in epilepsy diaries, repeated inter-episode intervals in mood disorders, fixed-lag repetitions in behavioural logs — a pooled or i.i.d. null confounds between-subject heterogeneity with within-subject structure, and a within-subject permutation null conditional on each subject’s observed multiset isolates the ordering question at the cost

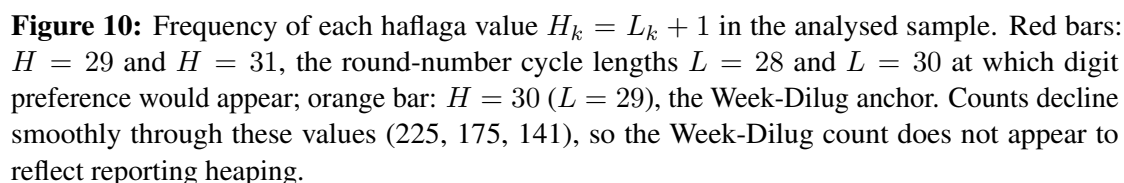
of answering only that question. The two should be reported together, with a simulation showing what the conditional test can detect. Only within-subject data with a defined generative alternative could sharpen this further; a parametric null from a fitted state-space model (Bortot et al., 2010) would be a natural next step.

5.7 Limitations

Coding and measurement. — Counts are invariant to the $H_k = L_k + 1$ relabelling provided the Week-Dilug anchor is shifted accordingly (verified: 26 Haflaga, 18 Week-Dilug under both conventions; computing on L_k with an unshifted anchor of 30 gives 12 Week-Dilug events). The haflaga distribution declines smoothly through the anchor ($H = 29$: 225 cycles; 30: 175; 31: 141; Figure 10), so the Week-Dilug count does not appear to reflect digit preference; no formal test of heaping was performed. Onset time of day is not recorded, so some events may be misclassified relative to the halachic requirement of a common onset period; a shift that is constant within a woman leaves her multiset and hence the H_W analysis unchanged, but shifts that vary within a woman are not addressed.

Scope. — Calendar dates are absent, precluding the 10 date-dependent rule types; body-sign patterns (*vesset haguf*) are invisible to cycle-length data. The two repeating-cycle rules were not analysed after zero occurrences were observed (Section 3.1). The sample consists of non-Jewish, regularly cycling, married NFP users (trial eligibility 18–42 years; Fehring et al., 2013), a self-selected, cycle-aware group who may be more regular than the general population; some halachic authorities hold that regularities in this population need not transfer to observant Jewish women (Ross, 2022b). Results should not be generalised without replication in the target population.

Statistical. — H_W tests exchangeability of each woman’s observed values and nothing more general. The simulations (Section 4.5) show high power against the simulated exact-repetition mechanism and against simulated whole-day AR(1) dependence of moderate strength ($\rho \geq 0.5$), about 50% power at $\rho = 0.25$, and, necessarily, none against dependence that leaves the distribution of these exact count statistics unchanged; the non-rejection should be read with those limits. The aggregate-count estimand weights women by follow-up; the truncation and



We applied three exchangeability nulls to the counts of five rule-defined menstrual patterns in 1,554 cycles from 118 women. Under population-level nulls the count vector is far in the tail, driven by Haflaga events at 2.3 times their global-null expectation (26 observed versus 11.3 expected; $z = 4.50$); under a within-woman permutation null conditional on each woman's own cycle-length values it is not, and all counts lie at or below their conditional means. Simulation shows type-I error at or below the nominal level for the marginal and Holm procedures and approximately nominal for the joint test; under the simulated alternatives the within-woman tests have high power against exact repetition and against AR(1) dependence of moderate strength (about 50% at $\rho = 0.25$, near-certain at $\rho \geq 0.5$); and the conclusions are insensitive to unequal follow-up. The results are consistent with between-woman heterogeneity in regularity generating the global excess and provide no evidence that within-woman ordering adds rule-defined events;

they do not establish the absence of serial dependence, and they say nothing about the forecasting value of the rules, which is the subject of ongoing work. To our knowledge this is the first formal randomisation-based test of the halachic vessel rules on empirical data. The general lesson is that for rule-defined counts in clustered longitudinal sequences the choice of exchangeability null is the analysis, and that a within-subject conditional null should be reported alongside any pooled one.

Ethics Statement

This study is a secondary analysis of a publicly archived, anonymised dataset (Fehring, 2012) collected in the randomised trial of Fehring et al. (2013). The original data collection was approved by the Institutional Review Board at Marquette University. No new data were collected, and no individual participant can be identified from the released dataset. Separate ethics approval for secondary analysis was not required under the applicable regulations.

Patient Consent Statement

No new data were collected and there was no contact with participants. According to the archive record of the dataset (Fehring, 2012), the human-subject data were anonymised for dissemination and data reuse was agreed to by the participants in the original consent form.

Conflict of Interest

The author declares no conflicts of interest.

Data Availability Statement

The source dataset is publicly available from Marquette University e-Publications (Fehring, 2012). All analyses reported in this article are reproducible from the released filtered dataset and analysis code at <https://github.com/dvirross/VesselVectorTest>: the filtered data file (`data/FilteredData.csv`; SHA-256 fingerprint `508ee88e...691f7811`, given in full in the Supporting Information and the repository README so that readers can verify file integrity), the pattern-counting code with an equivalence test between two implementations, the null-model generators, the simulation validation, the post-processing script that produces

every reported number, the notebook and the standalone Figure 2 script that together produce all figures, and the cached Monte Carlo replicates ($B = 50,000$ per null) that allow all results to be reproduced without re-simulation.

The preprocessing step is also reproducible. The script `scripts/filter_raw.py` derives the filtered file from the archived source file (1,665 cycles, 159 women) by removing three women with duplicated cycle records, women with fewer than 5 recorded cycles, and one duplicated copy of one woman’s cycles, and verifies that its output is identical to the released filtered file; no cycle-length criterion is applied, all archived lengths already lying between 18 and 54 days. The copy of the archived source file used by the author (`data/RawData.csv`) and the original filtering notebook are included in the code release; the archive itself is available from the Marquette repository cited above. Women retained in the filtered file have between 5 and 45 consecutive recorded cycles, and every retained woman’s cycle numbering is contiguous.

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